Drishti presentation speech

510 spoken words. Approximately 4-5 minutes with short pauses. Follow slides 1-8. Appendix slides are available for questions.

Slide 1 Drishti (0:00-0:30)

Good morning everyone. Today, we are presenting Drishti, our driver drowsiness monitoring project. Our mentor is Avani Bhuva. Our team members are Dhvani Tandel, N103, Krishhma Vira, N117, and Srishtee Varule, N114. The project studies how eye closure and yawning can contribute to warning decisions over time. Our current contribution is a temporal decision engine, supported by controlled software checks.

Slide 2 Why timing matters (0:30-0:59)

A single image cannot reliably explain a driver's condition. Closed eyes may indicate a normal blink, while a longer closure may require attention. Similarly, an open mouth could mean talking or yawning. Drishti therefore examines how long a cue continues and what has happened recently. This temporal context helps the engine distinguish brief movements from sustained or repeated behavior.

Slide 3 Research context (0:59-1:27)

Our paper draws on eye aspect ratio methods, physiological PERCLOS research, and datasets such as YawDD and UTA-RLDD. These sources examine different tasks, including blink recognition, yawning, and drowsiness classification. Their results cannot directly establish our system's accuracy. The research comparison helps us identify suitable evaluation resources and the need to calibrate thresholds across different people.

Slide 4 System pipeline (1:27-1:58)

The intended system begins with a camera and facial landmarks. A feature extraction stage would calculate eye and mouth aspect ratios, called E-A-R and M-A-R. The implemented engine receives those two numbers. It filters recent observations, tracks events, and calculates a warning state. Camera acquisition, landmark extraction, and warning delivery still require integration, so the verified component is the temporal engine shown here.

Slide 5 Eye and mouth cues (1:58-2:33)

The engine uses a median filter over up to seven observations to reduce isolated fluctuations. For eye state, it uses two thresholds. Eyes close when filtered E-A-R reaches zero point two three or lower, and reopen at zero point two seven or higher. Between them, the previous state remains unchanged. This behavior is called hysteresis. A yawn requires sufficiently large mouth opening to continue for at least zero point six five seconds.

Slide 6 Warning policy (2:33-3:06)

The engine combines current closure duration with recent yawns and a rolling closed-eye sample ratio. It produces four operational states: alert, fatigued, drowsy, and critical. Sustained closure reaches the drowsy rule at one point five seconds and the critical rule at three seconds. Other combinations can also trigger warnings. A smoothed evidence score supports these decisions, but it is not a probability of drowsiness or a clinical diagnosis.

Slide 7 Functional verification (3:06-3:40)

We checked eight deterministic scenarios using synthetic inputs and a controlled clock. All eight passed their specified assertions. They cover baseline behavior, hysteresis, blink counting, sustained closure, brief mouth opening, repeated yawns, event expiry, and startup behavior. The graph shows one synthetic closure and reopening sequence. It demonstrates filtering delay and changes in warning state. These results verify selected software behavior. They do not establish detection accuracy on real drivers.

Slide 8 Next evaluation (3:40-4:11)

Our next step is to integrate the camera front end and evaluate with independently labeled recordings. Calibration and testing should use different subjects. We should measure missed events, false warnings per hour, detection delay, and processing latency. Drishti currently provides an explainable temporal decision policy. Real-world validation will determine how effectively that policy supports driver monitoring. Thank you. We welcome your questions.